

CSA

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, NAGPUR

Department of Computer Science & Engineering

CSL 301: Database Management Systems

Date: May 07, 2025 (WED)

End-SEM-EXAM

Duration: 3 hour

Semester – IV

Max. Marks: 50

Important Instructions:

- This is a closed book, closed notes examination.
- This question paper comprises a total of 5 questions.
- All the questions are compulsory.

Q1	A hotel owner wants to develop a database system to manage its operations. The system should capture the following requirements: (6M+4M) • The hotel has a unique hotel ID, name, location, star rating, and contact information. • Each hotel has multiple rooms, uniquely identified by a room number within that hotel. Rooms have attributes like room type (single/double/suite), price per night, capacity, and availability status. • Guests can book rooms. Each guest has a unique ID, name, contact details, and ID proof. • A booking is made by a guest for a specific room in a specific hotel, with check-in and check-out dates, and a booking status (confirmed, cancelled, completed). • Guests may request services during their stay (e.g., laundry, spa, room service). Each service request includes the service name, date/time, quantity, and is linked to a particular booking. • The hotel employs various staff, such as receptionists, housekeepers, and managers. Each staff member has a staff ID, name, role, and salary. Staffs are assigned to one hotel. • A staff member may report to a manager. a) Consider aforementioned information and construct a well annotated E-R diagram, with inclusion of all type of attributes. b) Consider the E-R diagram constructed in part (a) and map the ER diagram into relational model, also specify all the attributes present in each table with proper representation.	(6+4) 10M	CO1
Q2 (a)	Consider the relations obtained from above question and write SQL Query for following. (5M) - List the guests who have spent more than 5000 in total across all their bookings. - Display all staff members along with the name of the manager they report to. - Hotel with the highest total revenue from bookings. - Staff who manage the most subordinates. - Guests who used services costing more than twice the average. - Hotels with >50% rooms booked. - Get the second most booked room type across all hotels. - Show average payment amount per hotel location. - Show bookings where the total service charges exceed the room charge. - Display guests who had more than 3 stays each longer than 7 days.	5M	CO1 CO2

Q2 (b)	Consider the relational schema given below and write Relational Algebra expressions for the following task.(5M) Event (EventID, Title, Date, Type, VenueID) Organizer (OrganizerID, Name, Contact) Attendee (AttendeeID, Name, Email) Venue (VenueID, Name, Location, Capacity) Schedule (ScheduleID, EventID, StartTime, EndTime, Description) a) Retrieve names of organizers who have organized events at more than 3 different venues. b) Find all attendees who attended events at every venue located in 'New York'. c) List the titles of events that were organized by more than one organizer. d) Get the names and emails of attendees who registered for the highest number of events. e) Identify attendees who attended two or more events on the same date.																																		
Q3 (a)	Consider the relation R(K, L, M, N, O, P, Q, R) with the following functional dependencies. $K \rightarrow LM, L \rightarrow N, N \rightarrow OP, P \rightarrow Q, Q \rightarrow R$. The relation R is decomposed into the constituent relations according to the following two decomposition schemes. D1: R1 (K, L, M), R2 (L, N, O, P), R3 (P, Q, R) D2: R1 (K, L), R2 (L, N), R3 (N, O, P, Q, R) Apply the chase algorithm to check whether the decomposition scheme D1 and D2 are lossless or lossy decomposition. Also check whether each decomposition scheme is dependency preserving or not with proper procedure.	[4+4] 8M	CO2 CO3																																
Q3 (b)	- Consider the following functional dependency and find the highest normal form for the same. Date_of_Birth $\rightarrow$ Age Age $\rightarrow$ Eligibility Name $\rightarrow$ Roll_number Roll_number $\rightarrow$ Name Course_number $\rightarrow$ Course_name Course_number $\rightarrow$ Instructor (Roll_number, Course_number) $\rightarrow$ Grade - After the decomposition of the table as R1 (Roll_number, Name, Date_of_Birth, Age), What is the highest normal form of R1? Why?	[4+3] 7M	CO2 CO3																																
Q4	Use Precedence graph to check whether the given schedule S is conflict serializable or not. If yes, then determine all the possible serialized schedules. <table border="1"> <thead> <tr> <th>T1</th><th>T2</th><th>T3</th><th>T4</th></tr> </thead> <tbody> <tr> <td></td><td></td><td></td><td>R(A)</td></tr> <tr> <td></td><td>R(A)</td><td></td><td></td></tr> <tr> <td></td><td></td><td>R(A)</td><td></td></tr> <tr> <td></td><td></td><td></td><td></td></tr> <tr> <td></td><td>W(A)</td><td></td><td></td></tr> <tr> <td></td><td></td><td>R(B)</td><td></td></tr> <tr> <td>W(B)</td><td>W(B)</td><td></td><td></td></tr> </tbody> </table>	T1	T2	T3	T4				R(A)		R(A)					R(A)							W(A)					R(B)		W(B)	W(B)			6M	CO3 CO4
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expression A database system stores index structures using disk blocks of size 4096 bytes. Each key is 12 bytes, each Block pointer (used in internal nodes) is 8 bytes, and each record pointer (used in B+ tree leaf nodes) is 10 bytes. (1+1.5+1.5=4M) a) Calculate maximum order (d) of a B-Tree node and draw the node structure. b) Calculate the maximum order (m) of an internal node in a B+ Tree and draw the node structure. c) Calculate the maximum number of a leaf node of the B+ Tree and draw the node structure.	[1+ 1.5+ 1.5] 4M	CO4
A database system needs to sort a file stored in 250,000 disk blocks. The system has 1,000 buffer pages available. (1+2+2=5M) a) Calculate the number of initial runs that will be produced. b) Estimate the minimum number of passes required to fully sort the file using an optimal k-way merge strategy. c) Estimate the total number of disk I/O operations.	[1+ 2+2] 5M	CO4

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